Exp 7: Random Sampling and Sampling Distribution

Program:

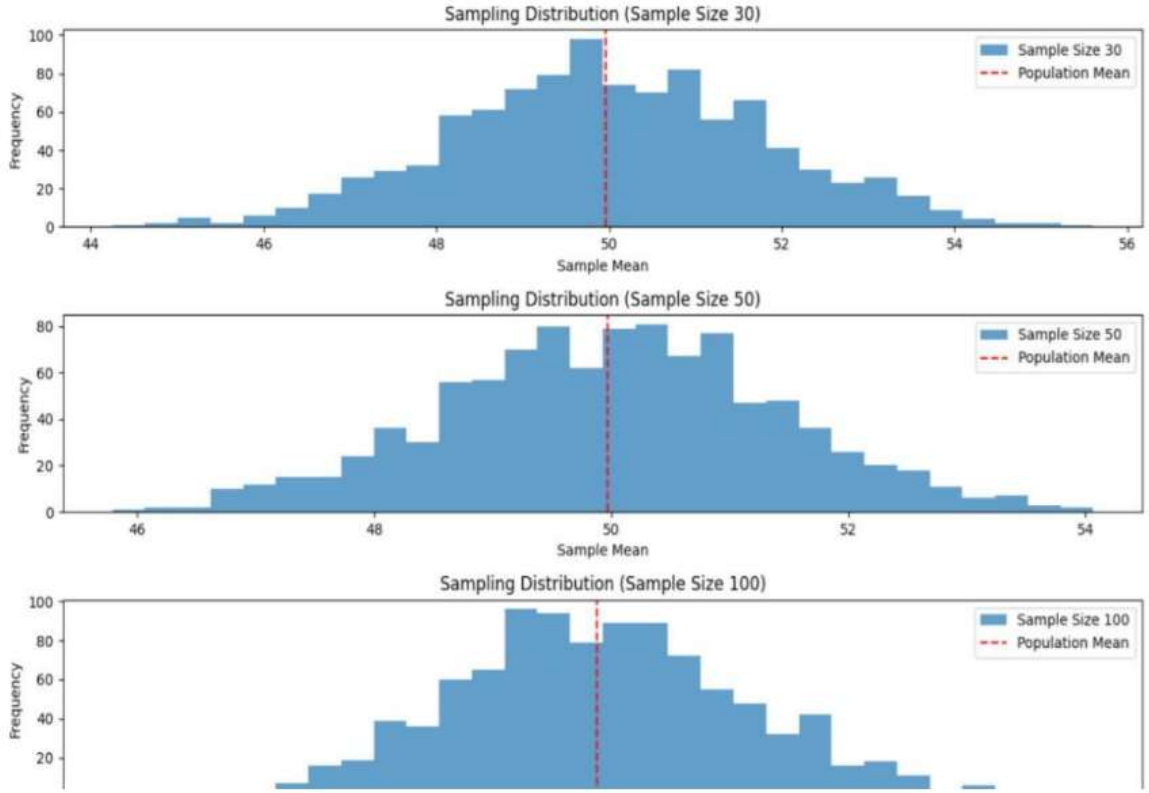
```
import numpy as np
import matplotlib.pyplot as plt
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)
sample_sizes = [30, 50, 100]
num_samples = 1000
26
sample_means = {}

for size in sample_sizes:
  sample_means[size] = []
  for _ in range(num_samples):
    sample = np.random.choice(population, size=size, replace=False)
    sample_means[size].append(np.mean(sample))
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
  plt.subplot(len(sample_sizes), 1, i+1)
  plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
  plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='PopulationMean')
  plt.title(f'Sampling Distribution (Sample Size {size})')
  plt.xlabel('Sample Mean')
  plt.ylabel('Frequency')
  plt.legend()

plt.tight_layout()
plt.show()
```

Output:


Sampling Distribution (Sample Size 30)
Frequency
Sample Mean
Sample Size 30
Population Mean
Sampling Distribution (Sample Size 50)
Frequency
Sample Mean
Sample Size 50
Population Mean
Sampling Distribution (Sample Size 100)
Frequency
Sample Size 100
Population Mean